GaitWay: A Modular Wearable System for Continuous Gait Monitoring and Quantitative Assessment of Parkinson's Disease

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Abstract

Impaired gait is a major debilitating symptom of neurological disorders such as Parkinson's disease. However, motor symptoms fluctuate throughout the day, so clinical visits often fail to capture these short-term changes. Assessing shorter-term gait performance could enable timely interventions to mitigate symptoms, such as optimizing the frequency and dosage of medications.

Existing clinical and consumer gait-tracking solutions are restricted to a single location, only capture limited data that lack sufficient fidelity, or suffer from prohibitively high costs. In addition, current designs often fail to resolve patient concerns regarding long-term comfort and the stigma associated with highly visible devices. There is a significant need to develop portable, affordable, and patient-friendly wearable sensors to monitor gait in a clinically meaningful way.

GaitWay is a system that incorporates sensors on the wrists & ankles, and a mobile app for real-time, accurate gait analysis. The modular design adapts to the patient's environment, where users can wear the sensors as strap-on bracelets or sensor-integrated socks. This flexible approach significantly improves aesthetic appeal and wearability, offering patients greater convenience and discretion. *GaitWay* offers a more competitive price, direct consumer access, and is specifically designed for patients.

Introduction

Gait impairment is a key symptom in neurological and neurodegenerative diseases, including Parkinson's, Huntington's, multiple sclerosis, and post-stroke motor deficits [1-2]. These impairments fluctuate throughout the day based on medication cycles, fatigue, and stress, and diminish patients' mobility as well as quality of life **(Figure 1)** [3-5]. Additionally, they can capture early changes and serve as sensitive indicators of disease progression [5-6]. Because motor symptoms fluctuate from hour to hour, a single measurement in a clinic fails to capture the variability patients experience in daily life [3-5].

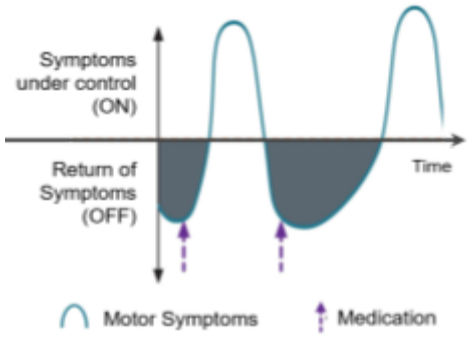


Figure 1. *Theoretical representation of short-term gait fluctuations throughout the day, motivating the need for continuous monitoring.*

However, current methods for assessing gait are limited to short clinical or lab visits utilizing motion capture technology [2,7]. Although traditional methods may offer stopgap solutions, they each have unique limitations. Camera-based lab systems are expensive, restricted to a specific location, and unrealistic for daily use, forcing physicians to rely on infrequent snapshots of patient motor function to adjust treatment plans, instead of a continuous stream of data [2,7]. While smartphones are more ubiquitous, they are unable to record multiple measurement points, therefore losing important spatial and temporal data needed to reconstruct gait with higher fidelity [8]. Commercial multi-sensor wearable devices are able to integrate multiple sensors in a more realistic platform for daily life, but are often costly and face practical barriers to broad adoption outside research and dedicated clinical programs [7]. An affordable, accessible, and accurate continuous gait monitoring system is needed to provide insight into short-term fluctuations in motor function.

GaitWay was designed as a wearable, multi-sensor platform for continuous gait assessment in naturalistic settings. By providing high-resolution, longitudinal data, this remote patient monitoring system can uncover new insights into an individual patient's motor function variability. In clinical care, such real-world data could enable earlier diagnosis, inform real-time interventions and personalized rehabilitation strategies, and support risk-stratification strategies to prevent serious accidents like falls or freezing of gait [6,9].

Beyond individual patient benefit, widespread use of *GaitWay* has the potential to accelerate clinical translation of therapeutics and advance the field of neurological research. *GaitWay* can enhance research reproducibility and support cross-study comparisons by standardizing gait data collection across diverse environments [4]. Post

Covid-19, the rise in home-based patient trials has increased the need for wearable, remote monitoring strategies to evaluate patient symptoms and provide insight into drug efficacy outside the clinic [10]. Ultimately, *GaitWay* has the potential to enhance translational research and reshape how clinicians and researchers understand mobility in neurodegenerative diseases and other conditions that affect motor function.

Materials and Methods

Sensing Modalities

To determine the optimal balance between accuracy, comfort, and efficiency, several sensing approaches were considered. This included inertial measurement units (IMUs), piezoresistive sensing elements for foot pressure measurement, and smart fabric-based integrations. IMUs were chosen to capture acceleration, angular velocity, and orientation with relatively low power consumption and compact hardware. Various design tradeoffs were explored, including sensor count, sensor placement, power architecture, charging strategies, and connectivity options.

Wearable Design

Mechanical form factors were explored to support different sensor placements and applications. Upper body sensors included watch-like devices, bracelets, and glove-based configurations, while foot-mounted designs included socks and anklets. These form factors were qualitatively evaluated for comfort, stability during motion, repeatability of sensor placement, and compatibility with daily activities and footwear. A sock was ultimately chosen for the ankle-mounted design, as the textile design of the smart sock improves comfort and wearability for the user, especially for outdoor use over long hours or in humid environments. Users are also given the option to attach the sensor using a strap-based anklet, allowing for flexibility in how the device can be worn depending on patient preference. For the upper body sensor configuration, a wrist-mounted strap, similar to commercial smartwatches and fitness trackers, was chosen since it provided a low-profile form factor, reducing stigma associated with wearing a medical device. Conversely, glove-based configurations were not chosen since gloves are not typically worn during everyday activities, so this could draw unwanted attention to the device. Additionally, glove-based systems require more effort to don and doff compared to a simple wrist-mounted approach.

Initially, sensors were attached to the exterior of a sock using Velcro to assess stability and signal quality during movement. Different attachment strategies were considered for their comfort, ease of donning/doffing, and heaviness on the device. Grip socks were ultimately chosen because their textured soles provide additional traction during walking, which may help reduce the risk of slipping and falls, a common concern for Parkinson’s disease patients. To protect sensors and standardize placement, 3D-printed

casings were designed using CAD software and fabricated from PLA filament (**Figure 2**). The initial enclosure designs included a hinged casing box and a snap-fit casing box. The snap-fit design provided a more secure enclosure and allowed for more sensor stability during movement, and was therefore selected for the final device housing design. After validation, the sensors were fully integrated into the textile to allow for comfortable, everyday wear, maintaining consistent alignment with anatomical landmarks.

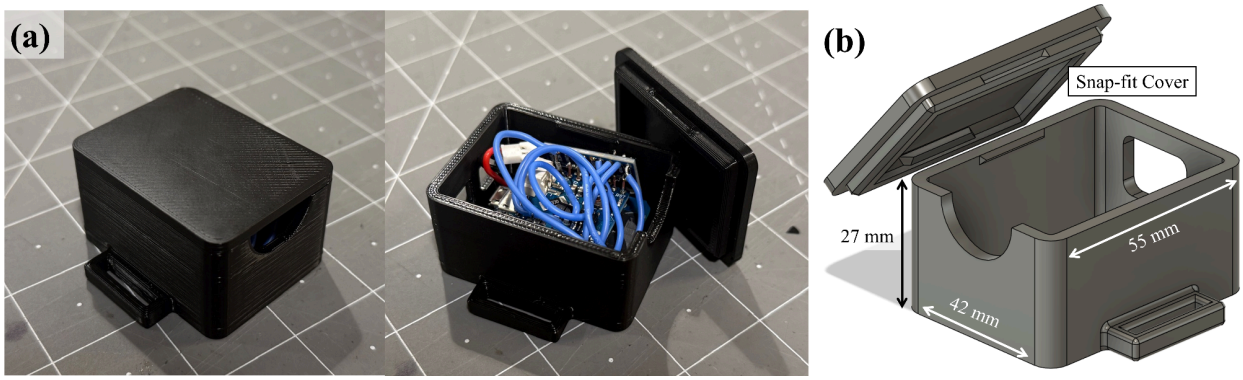


Figure 2 *(a) 3D-printed sensor casing (b) CAD model of snap-fit sensor casing*

Power and Connectivity

To determine how the system would be charged, two primary approaches were considered: a centralized battery pack mounted at the waist and wired to individual sensor nodes, and fully distributed systems in which each sensor module was powered independently. Distributed power systems improve modularity and reduce wiring but require synchronization across devices, adding another level of complexity in the software.

Connectivity was evaluated through wired and wireless communication. Wired configurations simplify synchronization and data integrity but reduce user mobility and increase mechanical complexity. Wireless approaches offer greater freedom of movement and scalability but introduce challenges related to latency, packet loss, power consumption, and synchronization across multiple sensor nodes.

Hardware Development and PCB Design

A simple wired prototype was constructed to validate signal quality from a single IMU connected to an ESP32 microcontroller. IMU breakout boards (MPU-9250) were used for initial tests. Wired connections were used to avoid potential synchronization issues inherent in wireless systems. Once the single-sensor signal was confirmed to be stable and responsive to movement, the system was scaled to multiple sensors to collect multi-point gait data. Signal quality, latency, and baseline noise levels were recorded and analyzed to ensure reproducibility. Materials for these experiments include ESP32

microcontrollers, MPU-9250 IMU sensors, wires, and a breadboard setup for rapid prototyping **(Figure 3a)**. These prototypes were then transitioned to a wireless device with similar components by incorporating an additional battery **(Figure 3b)**.

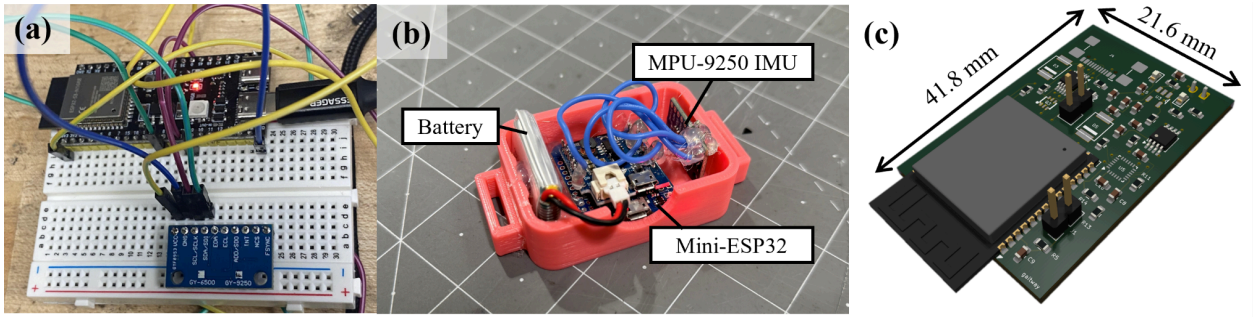


Figure 3 *(a) Wired prototype for initial testing (b) Wireless prototype with battery (c) PCB design reduced device size to 42 x 22 x 6mm.*

To make the system as lightweight and compact as possible, a printed circuit board (PCB) was designed to combine the IMU and ESP32, and removed unnecessary components of the mini-ESP32 board **(Figure 3c)**. A custom PCB reduced the device size to 42 x 22 x 6 mm. This seamless integration with the sock allowed the overall design to be small and inconspicuous.

Following validation of single-sensor operation, the system was expanded to a multi-sensor configuration, initially with two IMUs (one per ankle) and subsequently three IMUs (ankles and one wrist). Wireless synchronization protocols were assessed to ensure data alignment across sensors. Real-time acquisition and processing capabilities were verified, ensuring that multiple simultaneous streams of IMU data can be reliably captured and analyzed. This stage provided a robust platform capable of full-body gait monitoring.

Software Environment

To support real-time acquisition and processing of IMU data, Bluetooth Low-Energy (BLE) and WiFi connectivity were explored as options for wireless data transfer from multiple devices to a laptop. MATLAB and Python were also compared as two platforms for software development. Each platform was evaluated for ease of sensor integration, data streaming, calibration, drift compensation, and data visualization. The goal was to determine the most practical and reliable environment for processing multiple sensor inputs simultaneously. Sensor fusion algorithms were tested to combine accelerometer, gyroscope, and magnetometer data, ensuring accurate orientation and motion tracking. MATLAB and Python scripts were benchmarked for real-time processing and compatibility with ESP32.

Collected signals included raw acceleration, gyroscope, magnetometer, and foot pressure data, as well as derived kinematic features such as pitch angle and pitch velocity. These signals were used to compute gait parameters of interest for downstream analysis. Both machine learning based approaches and rules-based algorithms were explored to assess their suitability for gait characterization, classification, and robustness across system configurations.

System Validation Protocols

Raw sensor data acquired from the *GaitWay* prototype were compared with data collected from a commercially available wearable motion-tracking device. Healthy participants simultaneously wore both systems on the wrists and ankles while completing controlled walking trials. This protocol enabled direct comparison of acceleration and gyroscopic signals across devices, allowing assessment of signal consistency and temporal behavior.

To validate the extracted parameters, two distinct categories of walking experiments were designed and executed. In the initial phase, calculations were validated against known physical constants by strictly controlling each participant's timing and step length using a metronome and floor markings. This controlled environment, with both distance and duration predefined, provided a ground truth for the validation of speed calculations. Subsequently, the metronome frequency was systematically varied to alter the participants' walking velocity, facilitating a robust assessment of stride duration across different cadences.

The second experimental category focused on the differentiation between physiological and pathological gait patterns. Participants were first instructed to walk at a self-selected, natural pace and step length to establish a healthy baseline. They were then required to simulate Parkinsonian gait characteristics. These trials were utilized to evaluate whether gait asymmetry and double support time shifted in accordance with clinical expectations. Furthermore, the sensitivity of the sensor framework was tested by attempting to detect specific Parkinsonian symptoms, including reduced arm swing amplitude and the presence of tremors, during the simulated abnormal trials.

Final System Design

The final system design incorporates three wireless sensors, with one positioned on the wrist and two on the ankle. The mechanical design features a sock with integrated sensors, and a bracelet housing the electronics on the wrist. Moreover, the ankle design is modular, allowing users to decide between a sock or anklet. This design emphasizes comfort and wearability, allowing the user to keep the system on while performing daily activities.

Each individual sensor module consists of an IMU sensor and a mini ESP32 microcontroller, powered by a 400mAh Li-Pol battery. A printed circuit board containing each component was designed to keep the system compact, and also includes a voltage regulator and USB-C port. Data is transmitted wirelessly via Bluetooth Low-Energy, which is processed in 10 minute intervals to derive five gait parameters: stride length, stride duration, walking speed, left and right asymmetry, and double support.

Stride length refers to the distance covered between successive steps of the same foot and is often reduced in Parkinson's disease, reflecting hypokinesia and the characteristic shuffling gait; changes in stride length are also sensitive to medication state and disease progression. Stride duration, the time required to complete one full gait cycle, captures temporal control and rhythmicity of walking, which are commonly disrupted due to bradykinesia and impaired motor timing. Walking speed, defined as the rate of forward progression and determined by both stride length and cadence, serves as an integrated measure of overall mobility and functional status and is strongly associated with disease severity and fall risk. Left–right asymmetry quantifies differences in step length, timing, or stance between limbs and is especially relevant in Parkinson's disease, which typically presents with unilateral onset and persistent lateralized motor impairment. Finally, double support time, the portion of the gait cycle during which both feet are in contact with the ground, reflects balance confidence and postural stability; increases in double support are commonly observed in Parkinson's patients as a compensatory strategy to reduce instability and fall risk.

To compute a composite gait score, two machine learning models were used to independently characterize gait and arm swing during walking **(Figure 4)**. To analyze gait patterns, logistic regression, random forest, and XGBoost models were trained using gait data from individuals with Parkinson's disease and healthy controls, with the objective of predicting Parkinsonian gait based on extracted gait parameters. Arm swing is currently being quantified using a separate machine learning model, which will be combined with the gait model output to compute the final composite gait score.

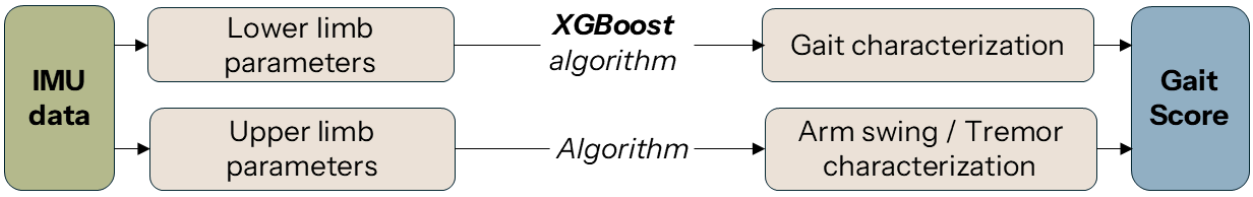


Figure 4 *Data processing pipeline to translate raw IMU data from sensors into an interpretable Gait Score using Machine Learning models.*

Key information about gait performance are reported to users through an iOS app, which displays the patient's most recent gait score, trends over time, and future gait predictions. **(Figure 5)** Clinical feedback highlighted two primary challenges in current Parkinson's management. First, inconsistent medication adherence often leaves clinicians without objective data regarding precise medication timings. Second, the fundamental objective of gait monitoring is to translate objective data into improvements in the patient's subjective quality of life. The app includes a feature for patients to log medication intake and input their subjective assessment of gait quality.



Figure 5 *Mobile app displaying information such as gait score, trends, predictions. The app's patient-facing interface also provides medication logs and subjective assessments of gait quality.*

Results

Mechanically, the *GaitWay* system can be comfortably mounted to the ankle or wrist for extended periods of time with no reports of irritation. The packaging is secure and ensures all electronics are able to be housed without moving. Additionally, each sensor unit is small enough to be concealed by pants or sleeves, creating a device that is wearable and inconspicuous.

Furthermore, using wrist and ankle-mounted IMU sensors during walking trials, the system reliably acquires real-time acceleration and gyroscopic data from multiple sensors through wireless data transmission **(Figure 6)**. The collected data is consistent

in magnitude, variability, and temporal behavior with measurements reported by commercially available motion sensor devices.

To translate raw sensor data into clinically relevant gait parameters, a rules-based algorithm was implemented. This algorithm enables consistent detection of heel strike and toe off events, which represent key temporal landmarks of the gait cycle. Although the machine learning algorithm determining gait score can only be fully validated in the future with clinical trials involving patients with Parkinson's disease, it suggests stable and reproducible performance across healthy subjects.

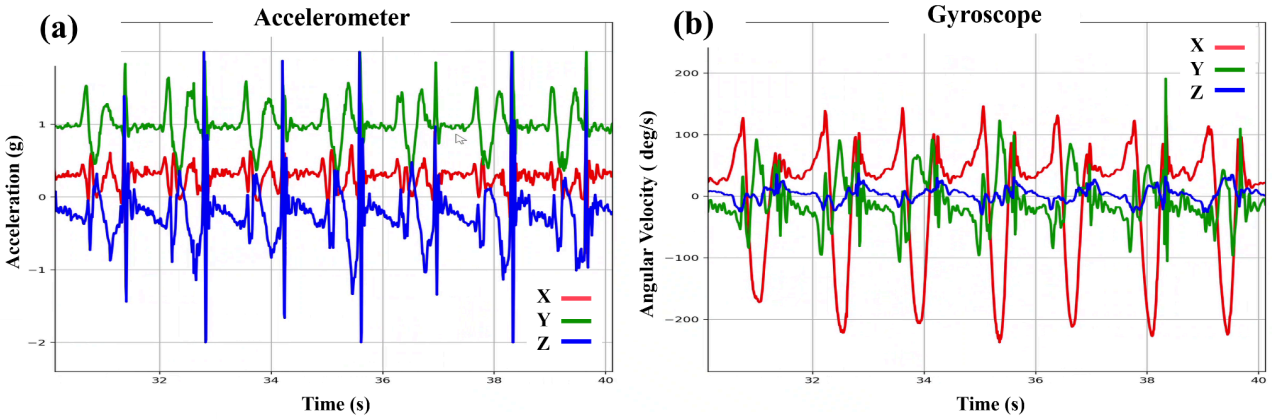


Figure 6 *(a) Accelerometer and (b) gyroscope signals from an ankle-worn GaitWay sensor during walking, with repeated patterns that reflect the cyclical nature of typical gait. Real-time data streaming was demonstrated, validating BLE as a viable option for wireless transmission.*

Ankle Motion and Rules-based Gait Event Detection

Real-time data acquired from the ankle sensors were processed to identify discrete gait events, including toe-off, mid-swing, and heel strike. To facilitate this, a rule-based algorithm was implemented to detect characteristic peaks and cyclical patterns within the raw sensor data. **(Figure 7)**

Mid-swing events were identified through the detection of positive signal peaks (blue dots). Subsequently, toe-off events were determined by identifying the local minima immediately preceding the mid-swing peak (green triangles), while heel strike events were defined by the local minima immediately following the mid-swing peak (red stars). As demonstrated by the consistent event localization across multiple trials, this approach provides a reliable method for the temporal characterization of the gait cycle.

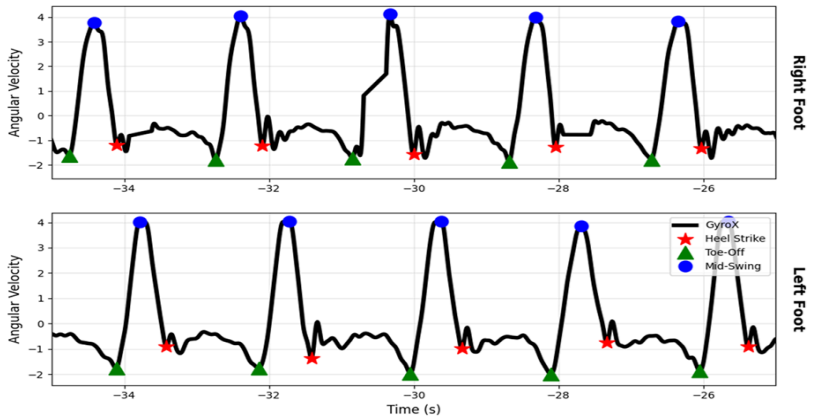


Figure 7 *Rules-based algorithm on Ankle GyroX data with consistent, reliable event detection.*

Wrist Motion and Frequency Domain Analysis

Wrist motion data were analyzed using a distinct methodology, as the raw signals do not yield discrete gait events comparable to those of the lower limbs. Given the periodic nature of the swinging motion, a Fourier transform was implemented to inspect the signal within the frequency domain. This allowed for the identification of a dominant frequency peak corresponding to individual steps. Consequently, cadence, stride duration, and arm swing amplitude were extracted from this spectral data.

The frequency-domain approach was validated by comparing wrist-derived stride durations against those obtained from ankle sensors. The "one-foot-one-second" trial served as the ground truth. As illustrated in the box plot, the estimates from both sensor locations demonstrated high levels of agreement with each other and with the expected 2-second duration (red dotted line).

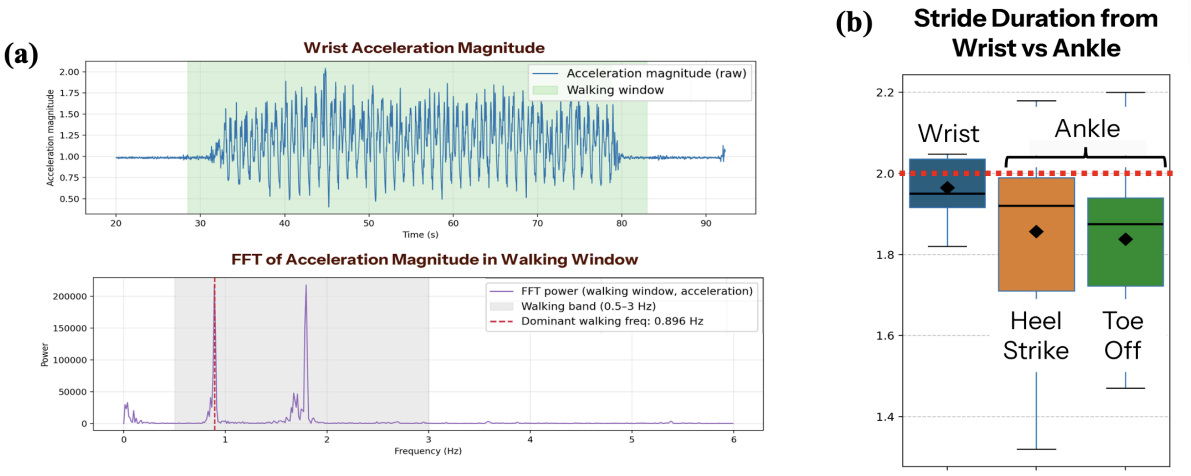


Figure 8 *(a) Frequency Domain analysis of the wrist sensor identified dominant walking frequency corresponding to expected stride duration. (b) Box plots comparing stride*

duration obtained from wrist and ankle sensors showed comparable results between both modalities.

Further validation of the stride duration parameter was conducted under two controlled conditions: a fixed pace and a decreasing pace. **(Figure 9)** In the variable pace trial, an increase in stride time was observed as walking speed decreased, while the constant rate trial produced uniform stride duration data at the expected 2-second interval.

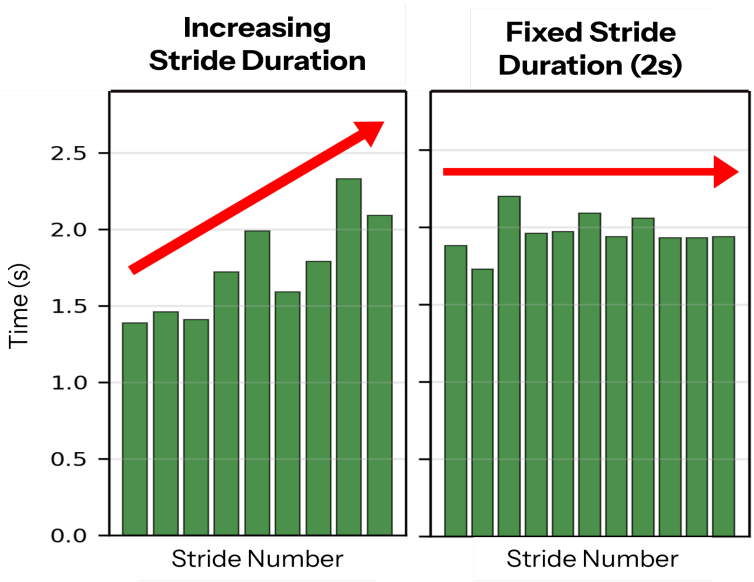


Figure 9 *Comparing a trial with an increasing stride duration (decreasing metronome pace) against fixed stride duration.*

In the experimental trials, participants in the healthy group walked with a fixed stride duration and natural rhythmic motion. Conversely, simulated Parkinsonian gait was characterized by slower walking speeds, asymmetric steps, and a significant reduction in arm motion.

Specifically, stride duration in the healthy trials remained consistent at 2 seconds, while it was significantly lower in the abnormal trials. **(Figure 10)** Furthermore, a reduction in arm swing amplitude was recorded in the abnormal trials, consistent with the limited movement observed during visual monitoring. **(Figure 10)**

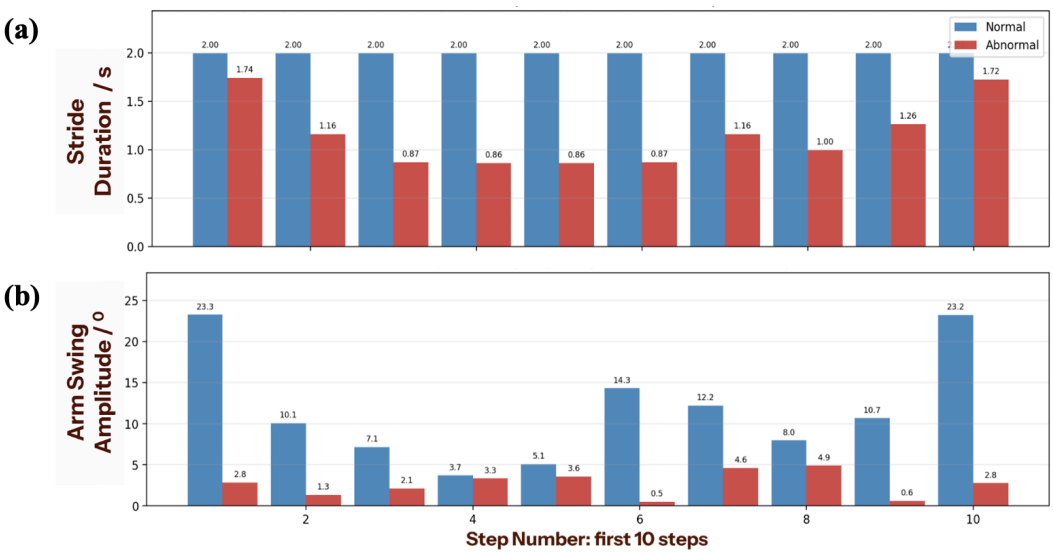


Figure 10 *(a) Decreased stride duration and (b) arm swing amplitude for simulated Parkinsonian gait.*

Gait stability was assessed through the measurement of double support time, where healthy gait is represented by blue indicators and abnormal gait by red. A general increase in mean double support duration was observed in the abnormal gait trials, alongside an increase in the double support percentage, calculated as a fraction of total stride duration. Higher double support times are indicative of gait instability or shuffling and are characteristic of the gait patterns observed in patients with Parkinson's disease.

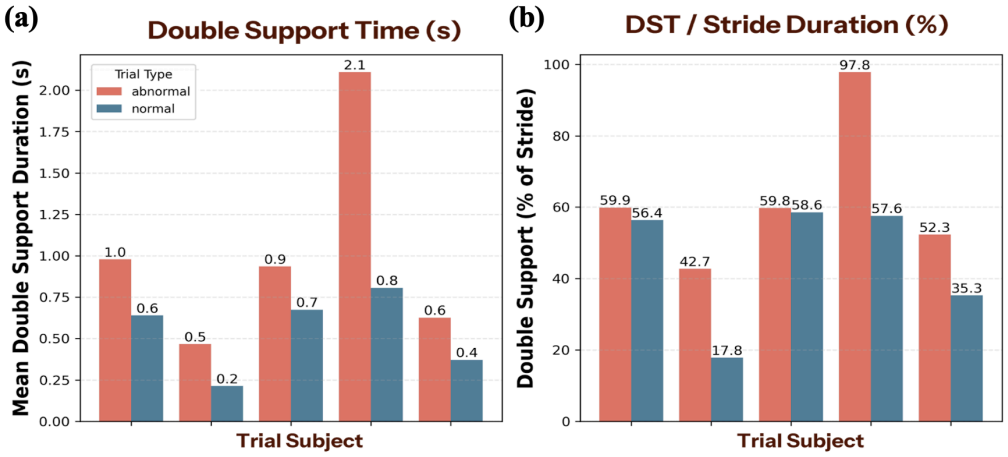


Figure 11 *(a) Decreased stride duration and (b) arm swing amplitude for simulated Parkinsonian gait.*

Definitive detection of pathological tremors cannot yet be claimed, as clinical validation on real patients is currently limited by a lack of access to clinical data. However, the system's ability to detect deviations from normal walking under simulated conditions was confirmed. In the healthy trials, zero tremors were detected, whereas eight distinct episodes were identified in the abnormal trials, resulting in a total tremor duration of 27 seconds. **(Figure 10)** These results suggest that the methodology is sensitive to deviations from healthy walking in a controlled environment.

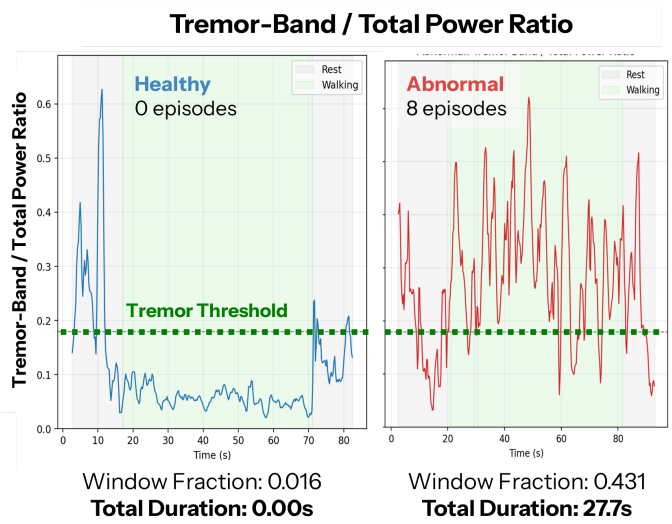


Figure 12 *No tremor episodes were detected in the healthy trial, compared to 8 episodes and a total tremor duration of 27 seconds in the simulated Parkinsonian trial.*

Model performance was compared utilizing the Area Under the Receiver Operating Characteristic (AUROC) curve, which illustrates the relationship between the true positive rate and the false positive rate. Within this framework, the XGBoost model demonstrated superior predictive capability, correctly assigning a higher probability of Parkinson's disease sample in approximately 77% of instances when compared to healthy controls. **(Figure 13a)**

Upon examination of the distribution of probability outputs, XGBoost exhibited the most distinct separation between healthy individuals and Parkinson's patients based on the extracted gait features. **(Figure 13b)** Based on this evaluation, XGBoost was selected as the final model for Parkinsonian gait prediction.

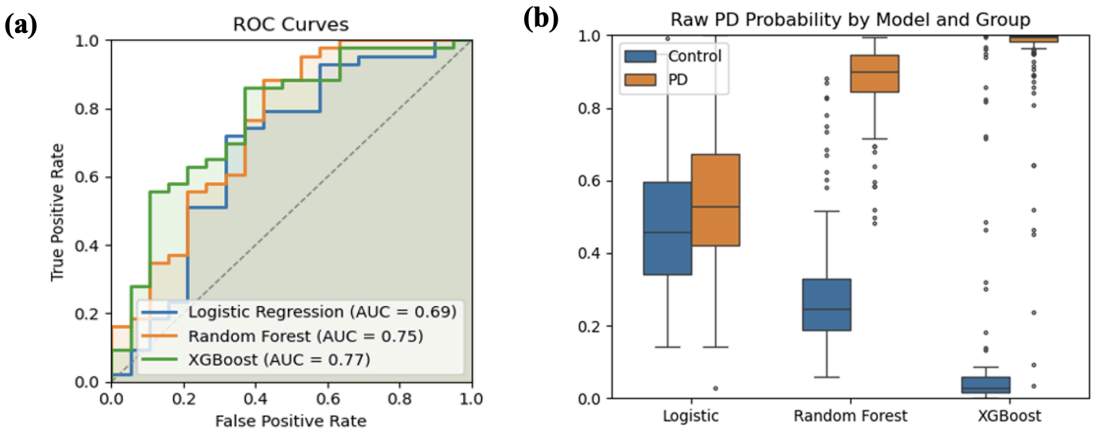


Figure 13 *(a) Highest area under ROC curve of 77% (greatest probability of assigning higher score to a PD patient as compared to a healthy subject). (b) Clearest separation in distribution of predicted PD probabilities for PD patients compared to healthy subjects.*

Discussion

GaitWay is designed as a wearable, multi-sensor platform for continuous gait assessment in real-life settings. Providing high-resolution and longitudinal data, this remote patient monitoring system can uncover new insights into an individual patient's motor function variability. In clinical care, such real-world data could enable earlier diagnosis, inform real-time interventions and personalized rehabilitation strategies, and support risk-stratification strategies to prevent serious accidents like falls or freezing of gait.

Beyond individual patient benefit, widespread use of *GaitWay* can accelerate clinical translation of therapeutics and advance the field of neurological research. *GaitWay* can enhance research reproducibility and support cross-study comparisons by standardizing gait data collection across diverse environments. Ultimately, *GaitWay* has the potential to enhance translational research and reshape how clinicians and researchers understand mobility in neurodegenerative diseases and other conditions that affect motor function.

GaitWay integrates a multi-sensor wearable platform into strap-on bracelets and textile socks instead of conventional ankle bands or shoe inserts. Although smart socks have been developed for runners, such systems can be prohibitively expensive for most consumers. Current systems on the market can range from $520 to $4800 [38], which significantly limits accessibility outside of clinical or research environments. In contrast, the estimated cost of the *GaitWay* prototype is under $100, making the system substantially more accessible for everyday use. *GaitWay* focuses on affordability, ease

of use, and direct access to consumers, as compared to current lab-based gait monitoring systems, which require a prescription to purchase or access to research labs.

Decreasing the barrier to access for such wearable technology is important, as it promotes a culture of self-monitoring in the healthcare system and a more patient-centric approach towards managing chronic neurodegenerative conditions. Patients can track their mobility changes in real-time and better understand how to optimize their medication schedules. They can also share this data with their clinicians and caretakers, allowing for more proactive and collaborative symptom management.

A limitation of the current system is a lack of access to patient data due to regulatory constraints. This means the system evaluation was limited to healthy subjects, simulated gait differences, and publicly available datasets. The current prototype also has an approximate battery life of about three hours of continuous use based on the expected Bluetooth Low Energy power consumption of the ESP32. Future iterations may explore power optimization strategies such as light or deep sleep modes to extend operational time for longer monitoring periods. Additionally, the current device uses a rigid PCB, which may limit comfort and flexibility during extended wear. Future work will explore the use of flexible PCB designs to improve wearability and allow the device to better adapt to body movement. Future work will focus on obtaining regulatory approvals and establishing clinical partnerships to enable data collection in Parkinson's populations. This will allow the system to be validated under real-world conditions and monitor the disease progression over time.

By addressing some of the key patient concerns, this system can increase patient compliance, which is pivotal for wearable devices to reap the full benefits of continuous monitoring. Rather than separate incompatible devices that patients have to constantly switch between, *GaitWay* offers a modular platform that integrates multiple devices into a single system, enabling a seamless transition between indoors and outdoors without breaking the continuous data stream and alleviating the technological burden on patients. *GaitWay* is the gateway to greater independence and mobility for patients with Parkinson's, where devices adapt to patients, not patients to devices.

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